

1 Introduction

- Introduce the problem with discovering physical systems. Can use the same analogy in the paper of newton vs kepler or find a different one.
- Talk about how using data to find patterns is useful, but we are more interested in finding underlying equations that govern the dynamical systems.
- We can use discovered equation to make predictions in areas where we have never seen data.

2 How it works

- We want to find f for $\dot{x}(t) = f(x(t))$
- We start with n observations of our data $x(t)$ at time points $t_0, t_1, t_2, \dots, t_n$
- We might also have m variables.
- Construct a matrix X of shape $n \times m$, where each row is a snapshot in time, and each column is that state variable. We also compute $\dot{X}$
- We construct a feature library $\Theta(X)$ of shape $n \times p$ where p is the number of candidate functions.
- Candidation functions can consist of many things including $x, x^2, \sin(x), \cos(x), \dots$

3 Finish with examples, ex lotka volterra